

Use the formula for the sum of the first n integers and/or the formula for the sum of a geometric sequence to evaluate the sums in 20-29 or to write them in closed form.

5.2.29. $1 - 2 + 2^2 - 2^3 + \dots + (-1)^n 2^n$, where n is any positive integer.

Theorem of sum of a Geometric Sequence:

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}$$

$$\sum_{i=0}^n (-1)^i * 2^i = \sum_{i=0}^n (-2)^i$$

$$\frac{(-2)^{n+1} - 1}{(-2) - 1} = \frac{(-2)^{n+1} - 1}{-3}$$

Solution: $\sum_{i=0}^n (-2)^i = \frac{(-2)^{n+1} - 1}{-3}$